KUMARAGURU COLLEGE OF TECHNOLOGY

Department of Artificial Intelligence and Data Science

DATA SCIENCE AND VISUALIZATION (24ADI204)

WEEK 4 REPORT

EDA DEEP DIVE

Youth Smoking and Drug Consumption Analysis

DATA-VIZ HACK-A-BIT 2026–2027

Team 16

Team Member Name	Roll Number
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1. Week 4 Contribution

Analysed skewness and feature distributions.

2. Work Carried Out

During Week 4, my individual contribution focused on exploring the distributions of demographic, personality and drug-consumption features in the UCI Drug Consumption dataset. I reviewed the notebook visualizations and examined how values were spread across categories and numerical variables. I also considered skewness to understand whether numerical feature distributions were approximately symmetrical or showed asymmetry.

3. Exploratory Data Analysis Performed

- Univariate analysis: Used count plots to examine Gender, Age Group, Education Level and drug-consumption categories. Histograms were used to inspect the distributions of Nscore, Escore, Oscore, Ascore, Cscore, Impulsive and SS.
- Bivariate analysis: Compared Gender and Age Group with Alcohol consumption, examined Education Level against Alcohol consumption, and used boxplots to compare Nscore across Alcohol and Cannabis consumption levels.
- Relationship analysis: Used scatter plots for Nscore versus Escore and Nscore versus Oscore. A correlation heatmap was created to examine the strength and direction of relationships among personality scores.
- Skewness review: The notebook reports skewness values for the seven personality features. All absolute values are below 0.5, indicating that these features are close to symmetrical in this analysis.

4. Visual Analysis and Findings

The notebook recorded 1,885 observations. The Age Group count plot showed the largest groups were 18–24 (605 records) and 25–34 (565 records), followed by 35–44 (335), 45–54 (275), 55–64 (89) and 65+ (16). The Gender count was 1,006 Female and 879 Male.

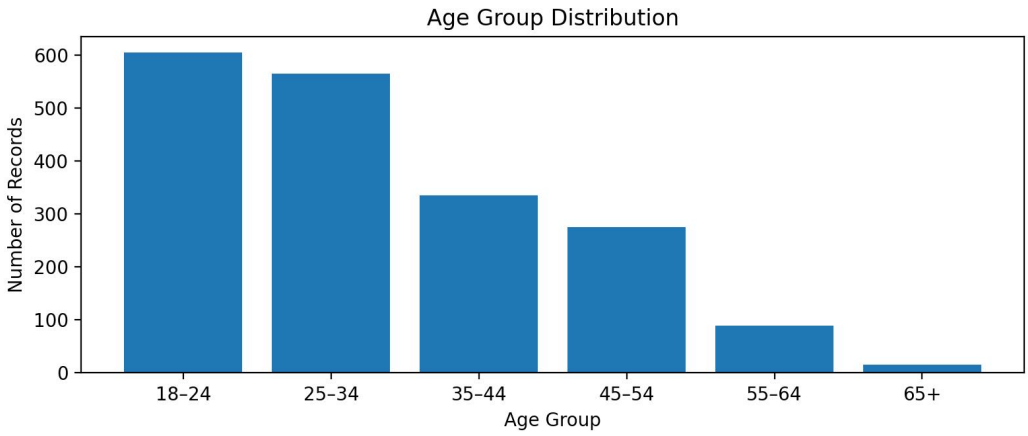


Figure 1: Age group distribution based on notebook value counts.

The skewness values were Nscore −0.0051, Escore −0.0017, Oscore −0.0103, Ascore −0.0039, Cscore −0.0051, Impulsive 0.0728 and SS −0.0407. These values are close to zero; therefore, no personality feature crossed the notebook's |skewness| > 0.5 selection threshold for Yeo-Johnson transformation.

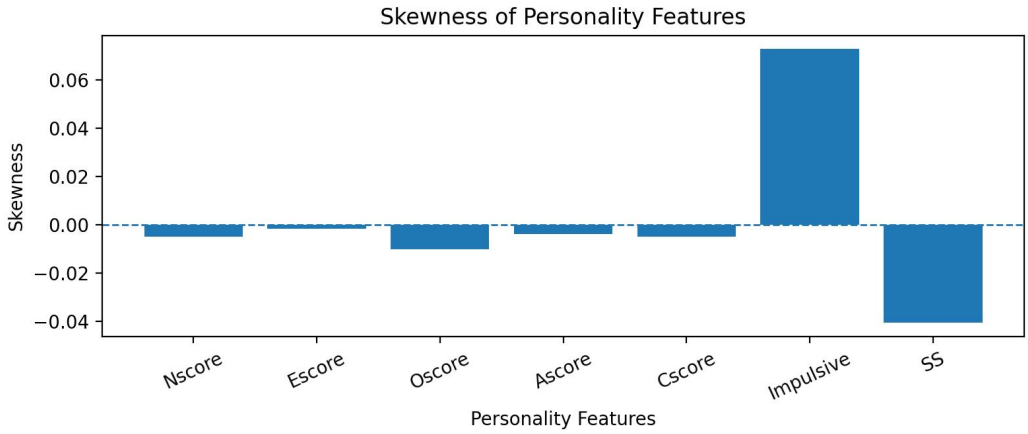


Figure 2: Skewness values of the seven personality features reported in the notebook.

The correlation heatmap showed the strongest positive relationship among the listed personality features was between Impulsive and SS (0.623). Nscore and Escore had a negative correlation (−0.414). The remaining relationships were generally weaker or moderate; the notebook did not identify any pair above the 0.80 threshold used in its later feature-selection analysis.

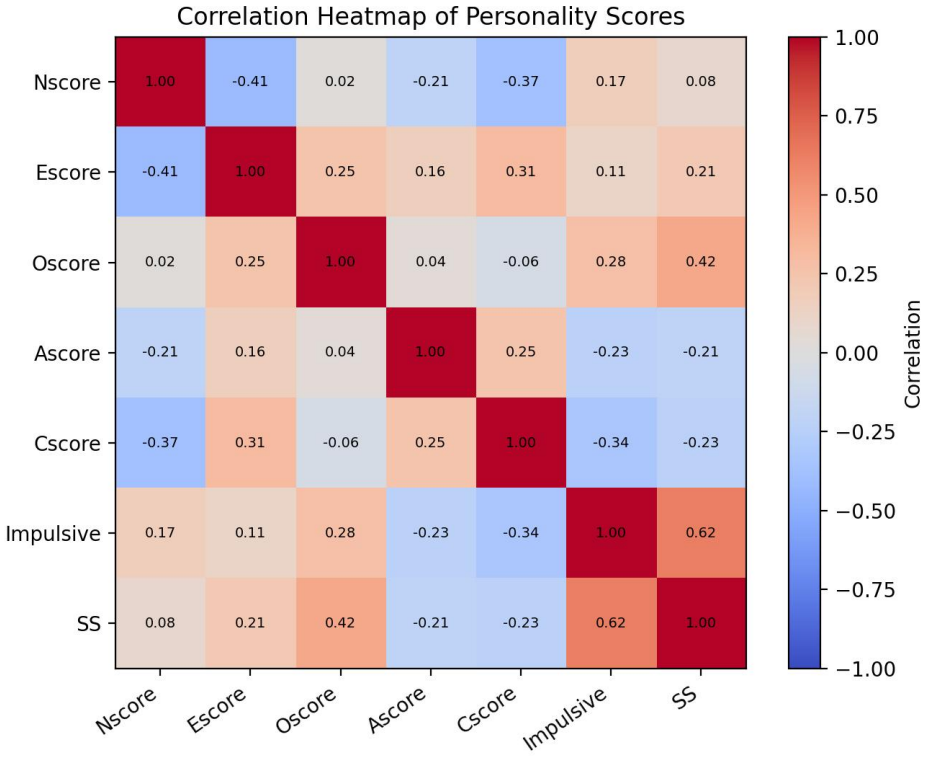


Figure 3: Correlation heatmap recreated using the correlation values displayed in the notebook.

5. Outcome of My Contribution

The EDA helped describe the distribution of individual features and explore relationships between demographic, personality and drug-consumption variables. Count plots and histograms supported univariate analysis, while grouped count plots, boxplots and scatter plots supported bivariate analysis. The correlation heatmap provided an overall view of relationships among personality scores. The notebook's final checks showed that the EDA dataset contained 1,885 rows and 35 columns, with zero missing values and zero duplicate rows. These observations provide a clearer understanding of the dataset for subsequent analysis.